

LEXIARAL

COMPLETE EDUCATIONAL APP STRUCTURE, FLOW, AND DEVELOPMENT PLAN

“Learn Words. Play. Grow.”

For Grade 3 Pupils under the English 3 ARAL Program

Research Title

“Strengthening Lexical Acquisition through Flashcard-Based Instruction and Digital Games: An Empirical Investigation of English Vocabulary Development among Grade 3 Pupils during the ARAL Program at Selected Elementary Schools in Rosario West District.”

1. App Overview and Research Basis

LEXIARAL is an educational vocabulary learning application designed for Grade 3 pupils under the ARAL (Academic Recovery and Accessible Learning) Program. It is intended to support learners who experience slow decoding, limited vocabulary, and weak reading comprehension.

The app combines the relevant elements of the Grade 3 English ARAL intervention into an interactive digital learning pathway. It particularly supports daily phonics and vocabulary drills, guided reading, comprehension practice, and progress monitoring.

2. Connection to the Grade 3 English ARAL Intervention

ARAL Intervention	LEXIARAL Feature	Purpose
Daily Phonics and Vocabulary Drill	Easy – Flashcard-Based Game	Vocabulary exposure, word recognition, repeated practice, and optional pronunciation support.
Guided Reading Sessions	Difficult – Story/Reading Challenge	Apply vocabulary knowledge while reading and understanding a passage.
Comprehension Worksheets	Digital Vocabulary and WH-Question Activities	Practice vocabulary and reading comprehension through interactive questions.
Peer Tutoring / Buddy Reading	Optional teacher/peer-guided use	Allow pupils to practice reading and vocabulary with a peer or teacher.
Weekly Progress Monitoring	My Progress, Scores, and Mastery Records	Track learner performance and progress.

3. General Objective

To strengthen English vocabulary acquisition among Grade 3 pupils by providing interactive, age-appropriate flashcard-based instruction and digital game activities that progress from word recognition to sentence understanding and story-based application.

4. Specific Learning Objectives

- Recognize and identify target English vocabulary words from the Grade 3 ARAL English materials.
- Connect target words with pictures and correct pronunciation.
- Understand the meaning of target vocabulary words when used in sentences.
- Use vocabulary knowledge to complete or interpret sentences.
- Apply vocabulary knowledge while reading short stories or ARAL-based passages.
- Answer vocabulary and WH-comprehension questions based on context.
- Monitor learning progress through scores, stars, completed levels, and vocabulary review.

5. Core Learning Progression

KNOW THE WORD → UNDERSTAND THE WORD → APPLY THE WORD IN CONTEXT → VOCABULARY MASTERY

The three levels intentionally increase cognitive and language demands:

- EASY: Word Recognition – Flashcards
- AVERAGE: Word Meaning and Usage – Sentences
- DIFFICULT: Vocabulary in Context – Stories and Reading Comprehension

6. Main App Flow

SPLASH SCREEN → WELCOME / START → HOME → LEVEL SELECTION → EASY → EASY RESULTS → AVERAGE → AVERAGE RESULTS → DIFFICULT → DIFFICULT RESULTS → FINAL RESULTS → PROGRESS / REVIEW / BADGES

7. Detailed Screen Structure

7.1 Splash / Welcome Screen

- LEXIARAL logo/title
- Tagline: “Learn Words. Play. Grow.”
- “English Vocabulary Game for Grade 3 Learners”
- START button

7.2 Home Screen

Main navigation:

- START LEARNING
- VOCABULARY REVIEW
- MY PROGRESS
- BADGES
- ABOUT

7.3 Level Selection

- LEVEL 1 – EASY: Learn the Words – Flashcard-Based Game
- LEVEL 2 – AVERAGE: Use the Words – Sentence-Based Game
- LEVEL 3 – DIFFICULT: Story Challenge – Story and Reading Comprehension

Recommended unlocking: Easy → Average → Difficult. The mastery requirement may be adjusted by the research team or teacher.

8. LEVEL 1 – EASY: FLASHCARD-BASED GAME

Purpose: Word recognition, vocabulary exposure, and repeated practice.

Flow: Instructions → Picture/Word → Question → Answer Choices → Answer Check → Feedback → Next Question → Final Score

8.1 Easy Level Activities

- Picture-to-Word: Select the word that matches the picture.
- Word-to-Picture: Select the picture that matches the target word.
- Matching: Match target vocabulary words with corresponding pictures.
- Listen-and-Choose: Listen to pronunciation and select the correct word.
- Flashcard Review: Tap a card to reveal the target word, picture, meaning, and pronunciation.

8.2 Easy Level Question Structure

Picture: [Actual ARAL vocabulary picture]

Question: “What is this?”

- A. [Choice]
- B. [Correct ARAL Word]
- C. [Choice]
- D. [Choice]

Correct feedback: “Great Job! +1 Star”

Incorrect feedback: “Nice Try! The correct answer is [WORD].”

8.3 Easy Level Completion

- Display final score and number of correct answers.
- Display stars earned.
- Provide positive feedback.
- REVIEW WORDS button.
- TRY AGAIN button.
- CONTINUE button.
- Unlock Average Level when the required mastery score is reached.

9. LEVEL 2 – AVERAGE: SENTENCE-BASED GAME

Purpose: Understand vocabulary meaning and usage in sentence context.

The same target words introduced in Easy should be reused in Average whenever appropriate. Difficulty increases because pupils must understand how a word functions in a sentence.

Flow: Instructions → Sentence → Missing/Target Word → Answer Choices → Answer Check → Feedback/Meaning → Next Question → Final Score

9.1 Average Level Activities

- Complete the Sentence: Choose the target word that correctly completes the sentence.
- Meaning in Context: Choose the meaning of a target word based on the sentence.
- Picture-Supported Sentence: Use a picture as additional support for understanding.
- Choose the Best Use: Select the sentence that uses the target word correctly.

9.2 Average Level Question Structure

Example format:

“The [target ARAL word] _____.”

Question: “Which word best completes the sentence?”

- A. [Choice]
- B. [Correct ARAL Word]
- C. [Choice]
- D. [Choice]

Feedback should briefly explain the correct answer using child-friendly English.

9.3 Average Level Completion

- Display final score.
- Display correct answers and stars earned.
- Provide encouraging feedback.
- Allow REVIEW WORDS or TRY AGAIN.
- Unlock Difficult Level when the required mastery score is reached.

10. LEVEL 3 – DIFFICULT: STORY / READING CHALLENGE

Purpose: Apply vocabulary knowledge while reading and understanding a story or short passage.

Stories should use or be appropriately adapted from approved Grade 3 ARAL reading passages when permitted. Target vocabulary should be clearly identified.

Flow: Instructions → Story Introduction → Read Story/Passage → Target Words → Vocabulary Questions → WH-Comprehension Questions → Feedback → Final Score

10.1 Story Screen Features

- Short, age-appropriate reading passage.
- Large and readable text.
- Target vocabulary words highlighted or visually marked.
- Optional audio for reading/pronunciation.
- Tap a target word to view its simple meaning and/or pronunciation.
- NEXT button to proceed to questions.

10.2 Difficult Level Question Types

- What does the target word mean in the story?
- Which sentence gives the correct meaning of the word?
- Who, What, Where, When, Why, and How questions.
- What happened before or after an event?

- Which detail from the story supports the answer?
- Which word best describes a character, object, or action?

10.3 Difficult Level Completion

- Display difficult-level score.
- Display vocabulary and comprehension performance.
- Give positive feedback.
- Allow REVIEW or TRY AGAIN.
- Proceed to Final Results.

11. Final Results Screen

After completing all three levels, display a summary:

- Easy Score
- Average Score
- Difficult Score
- Total Points / Stars
- Overall performance message
- Vocabulary Master badge, if all levels are completed

Buttons: REVIEW WORDS | PLAY AGAIN | HOME

12. Scoring and Feedback System

- Correct answer = +1 point/star.
- Incorrect answer = 0 points with supportive feedback.
- Show score after each level.
- Avoid harsh or discouraging messages.

Suggested performance messages:

- 90–100%: Excellent!
- 80–89%: Great Job!
- 70–79%: Good Work!
- Below 70%: Keep Practicing!

13. Vocabulary Review

Each learned word may contain:

- Target word
- Picture
- Simple meaning
- Example sentence
- Audio/pronunciation
- Level where the word was introduced

14. My Progress Dashboard

- Words Learned
- Easy Level Progress
- Average Level Progress
- Difficult Level Progress
- Total Stars
- Best Score
- Badges Earned
- Completed Activities

15. Badge / Achievement System

Badge	Requirement
First Word	Complete the first vocabulary activity.
Word Explorer	Complete Easy Level.
Sentence Builder	Complete Average Level.
Story Reader	Complete Difficult Level.
Vocabulary Master	Complete all three levels.

16. Vocabulary Content Structure

The vocabulary database must be based on the approved Grade 3 ARAL English 3 materials. Do not replace the ARAL target words with unrelated generic vocabulary. The same target words may be reused across the three levels so that the difficulty increases through the task.

ARAL Target Word	EASY	AVERAGE	DIFFICULT
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
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[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context
[Actual ARAL Word]	Picture + Word	Sentence	Story / Context

Important: The actual word list should be filled in only after the research team verifies the Grade 3 ARAL English 3 flashcards, modules, and reading passages.

17. One-Word Three-Level Model

- EASY – Introduce the target ARAL word using a picture, word, and optional pronunciation.
- AVERAGE – Reuse the same word in a short sentence and ask about its meaning or correct use.
- DIFFICULT – Reuse the same word in a story/passage and ask vocabulary and comprehension questions.

Learning progression: Word Recognition → Word Meaning and Usage → Vocabulary in Context.

18. Complete Navigation Flow

SPLASH SCREEN



WELCOME / START



HOME

— START LEARNING



LEVEL SELECTION



EASY



Instructions



Flashcard Game



Feedback



Score



AVERAGE UNLOCKED



AVERAGE



Instructions



Sentence Game



Feedback



Score



DIFFICULT UNLOCKED



DIFFICULT



Instructions



Story / Reading



Vocabulary Questions



WH-Comprehension Questions



Final Score



VOCABULARY REVIEW



MY PROGRESS



BADGES



ABOUT

19. Recommended App Screens

1. Splash / Welcome Screen
2. Home Screen
3. Level Selection
4. Easy Instructions
5. Easy Flashcard Game
6. Easy Results
7. Average Instructions
8. Average Sentence Game

9. Average Results
10. Difficult Instructions
11. Story / Reading Screen
12. Difficult Questions
13. Difficult Results
14. Final Results
15. Vocabulary Review
16. My Progress
17. Badges / Achievements
18. About the App

20. Suggested Technical / Development Components

- Buttons for navigation and answer choices.
- Image components for vocabulary pictures.
- Text labels for words, sentences, stories, scores, and feedback.
- Audio components for pronunciation and optional story reading.
- Score variable to track points.
- Question counter to track progress.
- Level status/unlock variable.
- Progress indicators.
- Local data storage for learner progress, if needed.
- Randomized question selection, if appropriate.
- Retry and review functions.

21. Design Guidelines for Grade 3 Pupils

- Use large, readable text and buttons.
- Use simple English instructions.
- Use child-friendly illustrations.
- Avoid crowded screens.
- Use positive feedback for both correct and incorrect answers.
- Provide audio support where possible.
- Keep stories short and age-appropriate.
- Use consistent icons and navigation buttons.
- Keep learning objectives more important than decorative game features.

22. Suggested App Development Roles

Role	Main Responsibility
Education / Research Team	Select and verify ARAL-aligned vocabulary, sentences, stories, objectives, and assessment content.
App Developer	Program screens, game logic, scoring, navigation, and data storage.
UI/UX Designer	Create child-friendly layouts, buttons, icons, and visual design.
Content Reviewer / Teacher	Check English accuracy, age appropriateness, and ARAL alignment.
Research Team	Conduct testing, collect results, monitor

23. Recommended Development Approach

- Finalize and verify the official ARAL Grade 3 vocabulary list.
- Map each target word to the Easy, Average, and Difficult levels.
- Prepare pictures and pronunciation resources for Easy.
- Write age-appropriate sentences for Average.
- Select or adapt approved ARAL reading passages for Difficult.
- Create vocabulary and comprehension questions.
- Build the app prototype.
- Test the app with teachers/research team members before pupil use.
- Revise the app based on feedback.
- Implement the app during the study and monitor learner progress.

24. Research Alignment

The combined app structure directly supports the research title and the Grade 3 English ARAL intervention. It transforms the existing flashcard and guided-reading components into a progressive digital learning experience. The three levels provide a clear sequence from recognizing vocabulary, to understanding vocabulary in sentences, and finally applying vocabulary within stories and reading comprehension.

25. Important Content and Implementation Notes

- Use actual Grade 3 ARAL English 3 vocabulary and reading materials as the primary content source.
- Do not invent a final vocabulary list before verifying the official ARAL materials.
- Use the same target vocabulary across levels when possible to demonstrate progressive learning.
- Stories should be short, age-appropriate, and aligned with approved ARAL reading materials when adaptation is permitted.
- The mastery threshold for unlocking levels should be agreed upon by the research team and adviser.
- The final app should be reviewed for content accuracy, usability, accessibility, and appropriateness for Grade 3 pupils.

26. Final Concept Summary

LEXIARAL is designed as a progressive vocabulary learning game for Grade 3 ARAL learners. The learner first encounters and recognizes ARAL vocabulary through flashcards in Easy Level. The learner then practices the same vocabulary in meaningful sentences in Average Level. Finally, the learner applies vocabulary knowledge in stories and comprehension tasks in Difficult Level. Scores, feedback, vocabulary review, badges, and progress monitoring provide motivation and a way to monitor learning.